

Definition

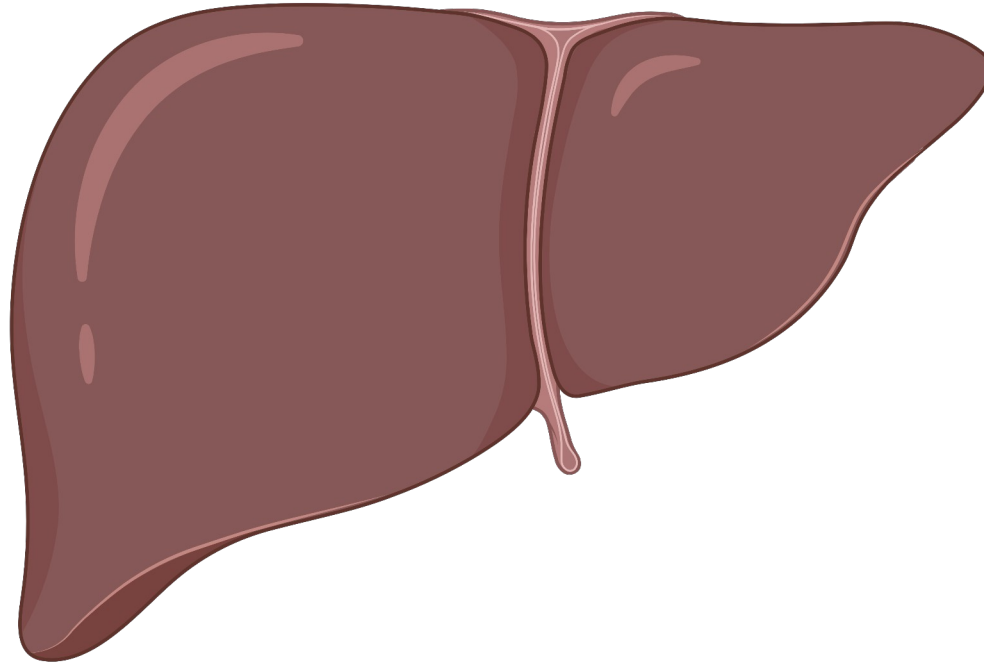
Etiologies

Management

Cases

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Acute Liver Failure



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DEFINITION AND CRITERIA FOR ALF

Must meet all three criteria

1

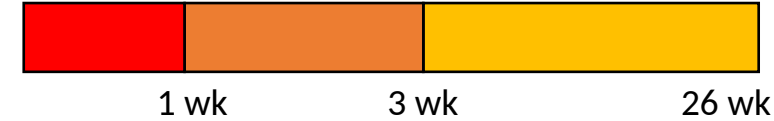
Acute Liver Inflammation

- Onset < 26 weeks
- ↑AST, ALT, and bilirubin
- No chronic liver disease
(exception HBV, AIH, & Wilson's)

Fulminant

Acute

Subacute



2

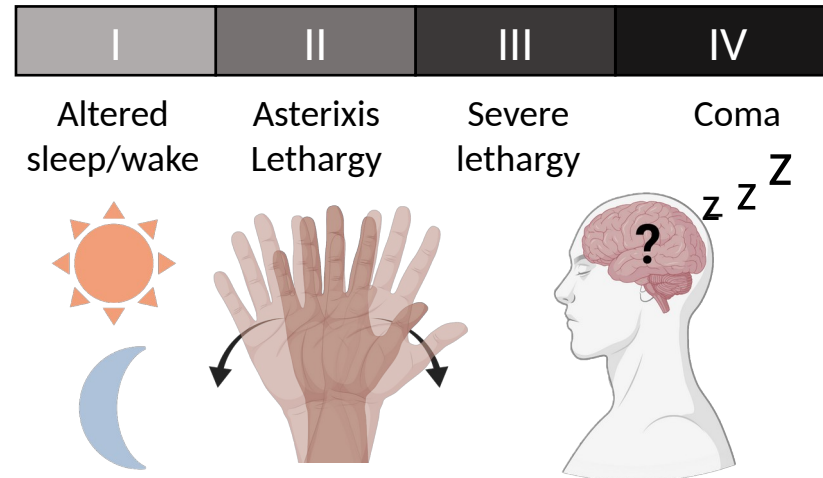
Coagulopathy (INR >1.5)

Marker of liver synthetic dysfunction

3

Hepatic Encephalopathy

Severity corresponds with risk of cerebral edema.



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Click to reveal
differential. Then click
again to reveal work-up

ETIOLOGIES AND EVALUATION

ETIOLOGIES

WORK-UP

Drugs/Toxins

- Acetaminophen
- Statins, AEDs, antibiotics
- Supplements
- Mushrooms (e.g., *Amanita*)

- APAP, salicylate level
- Toxicology screen
- Review NIH LiverTox

Viral

- Hepatitis A, B, D, & E
- Rarely: HSV, VZV

- Viral hepatitis serologies & PCRs
- HIV

Ischemia

- Budd-Chiari
- Shock liver / Ischemic hepatitis

- Liver US with doppler

Other

- Autoimmune Hepatitis
- Pregnancy-related (ie. HELLP, AFLP)
- Wilson's Disease

- ANA, anti-smooth muscle Ab, IgG
- Pregnancy test
- Ceruloplasmin, 24 hr Cu
- Liver biopsy

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MANAGEMENT OF ALF

1

Address
Complications

Cerebral Edema

Arterial ammonia
> 150

CT head, ICP monitoring
Intubation (airway protection)
↑ HOB, Hypertonics

Hypoglycemia

BG q1-2 hours
Continuous dextrose infusion

Multiorgan Failure

ICU monitoring, q4-6 hour labs
Intubation, vasopressors, RRT if needed

Coagulopathy

Reversal only if bleeding

Infection

± prophylactic/empiric antibiotics, antivirals

2

Give NAC

NAC improves liver transplant-free survival (even in non-APAP induced ALF)

3

Prognosticate

Use King's College Criteria (±MELD) to identify patients unlikely to recover spontaneously and more likely to benefit from a liver transplant

Bonus: Learn more about KCC

4

Transfer

Liver transplant centers allow for early transplant listing

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PROGNOSTICATION IN ALF (KING'S COLLEGE CRITERIA)

What is KCC?

Criteria developed to identify patients with ALF who are unlikely to recover spontaneously

When to use KCC

In APAP or non-APAP ALF to determine likelihood of requiring liver transplant.

How to interpret?

Meet criteria?

No

Spontaneous recovery more likely, continue supportive treatment

Yes

Consider early transfer to LT center and/or liver transplant

Other markers of poor prognosis

King's College Criteria

APAP toxicity	Non-APAP ALF
pH < 7.3	INR > 6.5 (PT > 100)
OR all of the below:	OR at least 3 of the following:
• INR > 6.5 (PT > 100) • Cr > 3.3 mg/dL • Grade III or IV HE	• Age < 10 or > 40 • HAV, HBV, drug-related ALF • Jaundice precedes HE by >7 d • INR > 3.5 (PT > 50) • Tbili > 18 mg/dL

↑ MELD

↑ end organ damage (↑ lactate, phosphate, Cr, Tbili)

↓ synthetic dysfunction (INR is most important prognostic marker!)

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29-year-old with recurrent pancreatitis presents with progressive abdominal pain x 4 d. They take 1000 mg of APAP q4h for pain. Over the past 24h, they developed nausea, vomiting, anorexia, and RUQ pain. Denies fevers, chills, or ill contacts.

Exam: Vital signs stable.

Alert and oriented x 4. No asterixis.

Epigastric and RUQ tenderness to palpation, but otherwise is unremarkable.

Labs: AST 9290, ALT 4550, ALP 118, total bilirubin 2.3, and INR 2.3.

What is the diagnosis?

Acute liver injury. Without encephalopathy, they do not meet criteria for ALF.

Which lab abnormalities represents synthetic dysfunction?

Elevated INR

Case 1

...On HD2, they are confused, oriented to self only. Exam shows asterixis.

What is the diagnosis?

Acute liver failure

Which is NOT part of the work-up for ALF?

A) Acetaminophen level

B) Liver US with doppler

C) Viral hepatitis serologies

Ethanol does not cause ALF

Which of the following is the best next step in evaluation and management?

NAC improves LT-free survival

B) Liver US with doppler

C) Heparin infusion

D) Obtain echocardiogram

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Case 2

35-year-old presents with acute liver failure due to fulminant hepatitis B. Initially, they had grade 1 HE. CT head was normal.

On HD2, they develop new nausea and vomiting not responsive to antiemetics.

VS: HR 58, BP 160/88, RR 20, SpO2 94% on 2L NC.

Exam: Somnolent but arousable, oriented to self, mild asterixis. No focal neurologic deficits.

Labs: Glucose 180. ABG 7.2/54/80/23. Ammonia 202.

What is the next step in evaluation?

Repeat **CT head** to evaluate for ↑ ICP. Ammonia > 200 predicts intracranial hypertension.

...CT head shows early cerebral edema without midline shift or herniation.

What is the next step in management?

- Elevate head of bed >30°
- Consider direct ICP monitoring
- Q1h neuro checks
- Hypertonics (hypertonic saline or mannitol)

Case 3

69-year-old with PMH of CHF, HTN, and type II DM is admitted with abdominal pain and fatigue.

VS: 37.5C, HR 145, BP **82/52**, T 36.5 °C

Exam: Moderate distress, cold distal extremities with 2+ pitting edema to bilateral knees. Oriented only to self, + asterixis.

Diagnostics:

- BMP: Cr 2.4 (baseline 0.9)
- ALT 3600, AST 5600, INR 2.2
- EKG shows sinus tachycardia. Troponin <0.04, BNP 2400. TTE shows LVEF of 10-20%.
- HAV IgM -, HBc IgM -, HBsAg -, HBs Ab +, HCV Ab -

What is the most likely cause of this patient's elevated LFTs?

A) Hepatitis C

B) Drug Overdose

C) Autoimmune hepatitis

↓ BPs likely suggest shock liver.

Which of the following is the most appropriate next step in management?

Reverse INR only if bleeding

Empirically treat only if febrile

C) Paracentesis

Vasopressors/inotropes

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Case 4

59 year-old with a history of ESRD s/p kidney transplant 5 years ago on chronic immunosuppression presents to a small community hospital with fever, chills, and ALF.

VS: T 38.8°C, HR 98, BP 117/58, RR 18, SpO2 95% on RA.

Exam: Jaundiced, lethargic, mild asterixis.

Labs: WBC 13K, AST 2300, ALT 1960, Tbili 22, INR 4.2, Cr 1.8 (unknown baseline).

APAP level and urine tox is negative.

What online resource can be used to determine if this patient's could have caused ALF?

The NIH website, **LiverTox**, can be used to review liver injury patterns and potential severity for most medications and supplements (livertox.nih.gov)

In addition to starting NAC, which of the following treatments is warranted while waiting for laboratory results?

A) Tylenol

Empiric acyclovir given pt febrile until HSV PCR returns

C) Prednisone

D) Activated Charcoal

What tools can be used to risk-stratify this patient and in what location should this patient be cared for?

King's College Criteria (or MELD) can help identify patients who are unlikely to spontaneously recover.

This patient meets non-APAP KCC criteria and is high risk for needing a liver transplant.

Transfer early to a liver transplant capable center.

King's College Criteria

Non-APAP toxicity

INR > 6.5 (PT > 100)

OR at least 3 of the following:

- Age < 10 or > 40
- HAV, HBV, drug-related ALF
- Jaundice precedes HE by >7 d
- INR > 3.5 (PT > 50)
- Tbili > 18 mg/dL

Currency Note & Framing Update (2026)

- Standing teaching verified current against AASLD acute liver failure practice guidance. The 3-criteria definition, etiology categories, "NAC for all ALF," King's College Criteria, and early transfer are unchanged.
- NAC nuance: the transplant-free survival benefit in NON-acetaminophen ALF is greatest in EARLY-stage disease with low-grade (I–II) hepatic encephalopathy; benefit is not established in advanced (grade III–IV) HE (Lee et al., Gastroenterology 2009). Still give NAC empirically to all ALF.
- Cerebral edema risk: arterial ammonia >150–200 $\mu\text{mol/L}$ predicts intracranial hypertension and severe HE — a marker for heightened neuro-monitoring (consistent with the deck's ammonia >150 / >200 cues).
- Prognostication: MELD is increasingly used alongside (and in some centers instead of) King's College Criteria; both should be applied with hepatology input. KCC remains specific but not sensitive — do not use a negative KCC to defer transplant-center transfer.
- Acetaminophen remains the single most common cause of ALF in the US/Europe (~50%); alcohol alone does NOT cause ALF.

Take-Home Points

- ALF = acute liver injury (<26 wks, no chronic liver disease*) + coagulopathy (INR >1.5) + hepatic encephalopathy. All three are required; HE is what separates ALF from acute liver INJURY. (*exceptions: HBV, AIH, Wilson's.)
- Drugs/toxins (esp. acetaminophen) and viral hepatitis account for >50% of cases; also screen for ischemic/vascular, autoimmune, pregnancy-related, and Wilson's causes. Send the etiology work-up in parallel.
- Manage in the ICU: address complications (cerebral edema, hypoglycemia, multiorgan failure, coagulopathy, infection). Reverse coagulopathy ONLY for active bleeding — INR is a key prognostic marker.
- Start NAC early in all ALF — it improves transplant-free survival even in non-acetaminophen cases (greatest benefit in early/low-grade HE).
- Consider transplant candidacy early and TRANSFER to a liver-transplant center as soon as possible — especially if King's College Criteria are met — before progression precludes safe transfer.

Attribution, License & Changelog

Adapted from the TeachIM original "Acute Liver Failure" (published April 2022).

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- License: Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Non-commercial educational use only.
- Optimization (June 2026): two surgical typo corrections — Case 2 slide "Acyclovir`1q1" → "Acyclovir"; Summary slide "Cerulopasmin" → "Ceruloplasmin". All original click-to-reveal animations and graphics preserved.
- Added: currency-note slide (NAC/HE-grade nuance, ammonia threshold, MELD vs KCC) + take-home slide. Companion files: ALF teaching guide (ALF-Learning-Guide-Final-Draft-4.3.docx), learner worksheet (ALF-learner-worksheet.pdf), completed board (ALF-teacher-worksheet-and-board-set-up.pdf) — reviewed, no changes needed.

Medical content should be clinician-reviewed before teaching.